A count is not an error bar: exact fluctuation floors for cell means of arithmetic fields

Youngrok Lee
Independent researcher, Republic of Korea

Abstract

Let $F(N)$ be an arithmetic field built by convolving a sign-valued multiplicative function against a fixed nonnegative weight — the model case is $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$, normalised as $Z(N) = C(N)/\sqrt{V(N)}$. Partition a band of N into cells indexed by divisibility, and ask whether a cell mean differs from the band mean. The usual error bar for that question is $\mathrm{sd}/\sqrt{n_c}$, built from the cell count n_c.

That error bar is wrong, and not by a constant. We give the exact fluctuation of any cell mean in closed form under the natural null in which the signs are replaced by independent ones (Lemma 1): three convolution terms, no simulation. We then show that this quantity does not decay like $n_c^{-1/2}$ but like $(\log N)^{-1/2}$ (Observation 4), because the summand $u_c(v)$ is a *coherent* sum rather than a self-averaging one: every N in the cell reads the same arithmetic input at the same v. Over a factor 128 in N, a count-based interval predicts the error bar shrinks by 11.3; it shrinks by 1.21, against the 1.19 that $(\log N)^{-1/2}$ predicts. An interval built from a count is therefore a factor 9.3 too narrow at the top of that range relative to the bottom, and in absolute terms the exact floor exceeds $\mathrm{sd}(Z)/\sqrt{n_c}$ by factors of 7.3 to 158.

The exact floor is not merely a corrected noise level. Under the label permutation that preserves every cell size while destroying the correspondence between cells and arithmetic (Lemma 7), the floor *collapses* by factors from 3.8 to 105: it is itself a second measurement of that correspondence, and quoting against it is conservative — by 3.8 at the cell that carries the effect, and by 105 at the cell that carries none. What predicts its size is the excess of the shift's singular series over same-cell pairs, a quantity that is scale-invariant by construction (Proposition 12) — so whatever produces the decay of the effect it measures, it is not the arithmetic that produces its size.

The conclusion applies to any cell mean of a field whose summands share a common arithmetic input, and is independent of the Goldbach context in which it was found.

1 Introduction

1.1 The setting

Let N range over the even integers of a band, let

$$C(N) = \sum_{n<N} \Lambda(n)\mu(N-n), \qquad V(N) = \sum_{v<N} \mu^2(v)\Lambda(N-v)^2, \qquad Z(N) = \frac{C(N)}{\sqrt{V(N)}}, \tag{1}$$

so that Z has second moment 1 under the null in which the $\mu(v)$ on their own support are replaced by independent signs; the exact asymptotic $V(N) \sim \prod_{q\nmid N}(1 - 1/(q(q-1)))\, N \log N$ is [2, Prop. 1], and the field (1) is the scalar to which the Huang–Li reduction of binary Goldbach [1] reduces. Partition the band into *cells* indexed by *depth* d, the number of $3, 5, 7, 11, 13$ dividing N. The question is whether $m_c - \overline{m}$, the difference between the cell mean of Z and the band mean, is larger than chance.

Answering it requires an error bar for $m_c - \overline{m}$. The reflex is $\mathrm{sd}(Z)/\sqrt{n_c}$ with n_c the cell count. This paper shows that reflex fails here, gives the correct quantity in closed form, and identifies the mechanism — which is generic, not specific to (1).

1.2 Results

- Lemma 1 gives $\mathrm{Var}(m_c - \overline{m})$ exactly under the sign null, as three convolution terms.
- Observation 4 identifies its decay: $(\log N)^{-1/2}$, not $n_c^{-1/2}$.
- Lemma 7 is the control that separates a property of the cell *sizes* from a property of the correspondence between cells and arithmetic.
- Measurement 9 runs it, and finds that the effect survives while the floor itself does not — so the floor is a second measurement of the same correspondence.
- Proposition 12 shows the quantity predicting the floor's size is scale-invariant.

1.3 What is not claimed

- **No decay exponent for the effect is quoted.** At the three shallowest depths the amplitude does not clear the exact floor at any scale measured, so nothing can be fitted there, and whether the exponents differ by depth is not decided by the accessible range.
- **No distributional law for Z is asserted.**
- **No consequence for the Goldbach problem.** Nothing measured here bounds $C(N)$ in either direction. No sweep over parameterisations is carried out, and none is claimed.
- **Observation 4 is offered for its form, not its constant.** Its heuristic is quantitatively loose by factors running from 2.49 at depth 0 to 6.20 at depth 5 — the 2.5 is the shallowest cell, not the range; the exponent is confirmed directly instead.

2 The floor, in closed form

Lemma 1 (exact cell moments)**.** *Let a band of even N be partitioned into cells, let c be a cell of size n_c inside a band of size n, let m_c and $\overline{m}$ be the means of Z over c and over the band, and set*

$$u_c(v) \;=\; \sum_{N\in c} \frac{\Lambda(N-v)}{\sqrt{V(N)}}, \qquad Q_{cd} \;=\; \sum_{v} \mu^2(v)\, u_c(v)\, u_d(v).$$

Then, when the $\mu(v)$ on the surviving support are replaced by independent signs,

$$\mathrm{Var}\big(m_c - \overline{m}\big) \;=\; \frac{Q_{cc}}{n_c^2} \;-\; \frac{2\,Q_{ca}}{n_c\, n} \;+\; \frac{Q_{aa}}{n^2}, \tag{2}$$

where a denotes the whole band. Every term is computable exactly by one convolution; no simulation is involved.

Proof. Under independent signs $E[\varepsilon(v)\varepsilon(v')] = \mu^2(v)\delta_{vv'}$, so $\mathrm{Cov}(n_c m_c, n_d m_d) = Q_{cd}$ exactly. Expanding $\mathrm{Var}(m_c - \overline{m})$ bilinearly gives the three terms. □

Note 2 (all three terms are needed)*.* The substitution $u_c(v) \approx n_c/\sqrt{V}$, which gives the right scale for Q_{cc}/n_c^2, is not specific to c; applied to all three terms it returns the same value S for each and hence $S - 2S + S = 0$. So it cannot be used to estimate the variance of the difference. Exactly, the variance is smaller than Q_{cc}/n_c^2 by a factor that is a structural constant rather than noise.

Measurement 3 (the closed form, checked against simulation)**.** The lemma does not need simulation, but simulation is the only thing that would catch an error in the formula, and every normalised cell mean below is divided by it. Run in the band $(10^5, 2 \cdot 10^5]$ with 2000 draws:

depth	0	1	2	3	4	5
n_c	19181	21097	8183	1421	115	3
closed form	0.140080	0.052724	0.167579	0.273630	0.378073	0.639330
MC/closed	0.9920	0.9993	0.9918	1.0044	1.0157	1.0320

— worst deviation 0.0320, at the depth whose cell holds three elements, and the six ratios straddle 1 three and three.

The six ratios are estimated from the *same* draws, so their deviations are strongly correlated and their common sign would be one observation, not six; that is why the check is run at 2000 draws rather than at a number where the Monte-Carlo precision, here $1/\sqrt{2 \cdot 1999}$, is comparable to the deviation being judged.

The structural constant of Note 2 is visible directly: at the top octave of $1.6 \cdot 10^7$, $\mathrm{Var}/(Q_{cc}/n_c^2)$ is 0.113119, 0.016302, 0.288567, 0.551516 at depths $0, 1, 3, 5$, and at a quarter of that N the three shallow ratios agree to six digits — 0.113118, 0.016303, 0.288571. Depth 5 does not: it reads 0.557654 there against 0.551516 at the top, differing in the third digit, and it is much the smallest cell — 266 values of N in the top octave against 1 687 911 at depth 1 (`lab_cellmom_montecarlo.py`).

3 The floor's uncertainty does not fall like a count

Observation 4 (coherent sums)**.** The error bar of Lemma 1 does *not* decay like $n_c^{-1/2}$. For fixed v, about $n_c/\log N$ of the terms of $u_c(v)$ are nonzero, each of size $\log N/\sqrt{V}$, so $u_c(v) \approx n_c/\sqrt{V}$ and

$$\frac{Q_{cc}}{n_c^2} \approx \frac{\sum_v \mu^2(v)}{V} \asymp \frac{1}{\log N},$$

independently of n_c; and by Note 2 the variance itself is a fixed fraction of this. So the standard error falls like $(\log N)^{-1/2}$.

The heuristic is quantitatively loose and is offered for the *form* and not the constant: at $N = 1.6 \cdot 10^7$,

depth	0	1	2	3	4	5
Q_{cc}/n_c^2	0.123577	0.121208	0.146026	0.184159	0.235171	0.307074

against the heuristic's $\big(\sum_{v<N} \mu^2(v)\big)/V(N) = 0.049540$. The row is not monotone in the depth; its minimum is 0.121208 at depth 1.

Measurement 5 (the exponent, fitted)**.** The form is confirmed directly. Fitting se $\propto N^{-b}$ to the exact floor across eight octaves gives $b = 0.039451, 0.039671, 0.039388$ at depths $2, 1, 0$, against the 0.5 that a count would give and against 0.036038, the value of $1/(2\langle \log N \rangle)$ with $\langle \log N \rangle$ the mean of $\log N$ over the eight octave midpoints. Nothing turns on the third decimal: the point is that 0.039 is near 0.036 and nowhere near 0.5 (`lab_cell_floor.py`).

Note 6 (a count is not an error bar)**.** The consequence is not confined to this field. The fit runs over the eight octaves from $(6.25 \cdot 10^4,\ 1.25 \cdot 10^5]$ to $(8 \cdot 10^6,\ 1.6 \cdot 10^7]$, and the abscissa is the octave midpoint — which is what makes $1/(2\langle \log N \rangle) = 0.036038$ come out, with $\langle \log N \rangle = 13.8744$. The factor in N is therefore $1.2 \cdot 10^7/93750 = 128$. Over it, $n_c^{-1/2}$ says the error bar shrinks by $\sqrt{128} = 11.3$; it shrinks by $128^{0.0395} = 1.21$, against the 1.19 that $(\log N)^{-1/2}$ predicts over the same factor. **An interval built from a count is therefore a factor** $11.3/1.21 = 9.3$

too narrow at the top of this range relative to the bottom, and in absolute terms the discrepancy is larger still: at the top octave the exact floor exceeds $\mathrm{sd}(Z)/\sqrt{n_c}$ by factors of 7.3 to 158, with $\mathrm{sd}(Z) = 0.924687$ over that octave — and 0.929384 over the whole fitted field, so the reading of sd barely moves the figures. The factor does *not* grow with the cell. What grows with the cell is the floor itself, 0.118 to 0.412 from depth 0 to depth 5, exactly as $Q_{cc}/n_c^2 \asymp 1/\log N$ predicts; the factor also carries $\sqrt{n_c}$, and n_c falls from $1.53 \cdot 10^6$ to 266 across those depths, which is the larger movement. By depth the factor reads 158, 62, 124, 84, 33, 7 (`audit_cellfloor_sdz.py`). The cause is that $u_c(v)$ is a coherent sum, not a self-averaging one, and the same is true of any cell mean of a field whose summands share a common arithmetic input.

4 The control, and what running it shows

Lemma 7 (the placebo key)**.** *Let the cells be indexed by a labelling $\ell(N)$, and let π be a permutation of the even integers in the band. Replacing ℓ by $\ell \circ \pi$ preserves every cell size and leaves the field Z pointwise unchanged, while destroying the correspondence between cells and arithmetic. Any cell-indexed statistic that survives this replacement is a property of the cell sizes and not of that correspondence.*

Proof. The permutation acts on labels only; the multiset $\{Z(N) : N \text{ in the band}\}$ and the multiset of cell sizes are both invariant, and the joint distribution of $(\ell \circ \pi, Z)$ is that of independent labels. $\square$

Measurement 8 (the cell means are not zero)**.** Against the exactly computed floor of Lemma 1, the cell means of the field (1) are not zero. Over the eight octaves from $(6.25 \cdot 10^4,\ 1.25 \cdot 10^5]$ to $(8 \cdot 10^6,\ 1.6 \cdot 10^7]$, $\max_c |z_c|$ reads

$$10.099,\ 10.954,\ 11.179,\ 12.968,\ 11.885,\ 11.026,\ 10.019,\ 9.064,$$

clearing Bonferroni in each. The signal is carried by the deep cells: at the top octave depths $3, 4, 5$ sit at $z = -1.55, -4.46, -9.06$ while depths $0, 1, 2$ sit at $+0.04, +0.69, -0.06$ (`lab_cell_floor.py`).

Measurement 9 (the effect survives the placebo; the floor does not)**.** Run on the octave $(2 \cdot 10^6,\ 4 \cdot 10^6]$, with cell sizes preserved exactly and the floor recomputed from scratch for each permutation:

depth	0	1	2	3	4	5
n_c	383617	421978	163568	28507	2263	67
z_c (true)	+0.1069	+1.1133	−0.2314	−2.3990	−5.9997	−11.0258

against $\max_c |z_c|$ over ten label permutations reading

$$0.9359,\ 1.6073,\ 1.9921,\ 1.5687,\ 2.1392,\ 1.1216,\ 1.7178,\ 1.1348,\ 3.2040,\ 1.3192,$$

mean 1.6741, never above 3.3. The correlation between depth and z_c is -0.9106 for the true labelling and $+0.0042$ on average under permutation. So what is detected is the correspondence between cells and divisibility, not the cell sizes.

The same run corrects a natural reading of Lemma 1. One expects the floor se_c to be a property of the cell sizes, hence nearly invariant under permutation. It is not: the floor **collapses**, by factors from 3.8 at depth 5 to 105 at depth 0 — $1.2400 \cdot 10^{-1}$ against $1.1794 \cdot 10^{-3}$ there (`lab_mask_placebo.py`).

Note 10 (the floor is itself signal)*.* The mechanism of that collapse is the three-term structure of (2). For a random subset of the band, $u_c(v) \approx (n_c/n)\,u_a(v)$, and then $Q_{cc}/n_c^2 - 2Q_{ca}/(n_c n) + Q_{aa}/n^2$ cancels to leading order, leaving only the fluctuation around proportionality. An arithmetic cell breaks that proportionality, and the floor is what is left over. **The exact floor is therefore not a noise level but a second measurement of the same correspondence**, and quoting z_c against it is conservative — but by how much depends on the cell, and the figure worth quoting is the one at the cell that carries the effect. At depth 5 the floor collapses by 3.8 under the permutation, so with the permuted floor in the denominator that cell would read $z \approx -42$ rather than -11: half an order of magnitude, not two. Two orders is depth 0, where the collapse is 105 and $z = +0.11$ — a cell with no effect to be conservative about. The conservative figure is the one kept throughout.

5 What predicts the floor's size

Write $\mathfrak{S}_2(h)$ for the Hardy–Littlewood singular series of the shift h — the local density of pairs $(p, p+h)$ — and, for a cell c in the band, let $E_{\text{same},c}[\mathfrak{S}_2]$ be the mean of $\mathfrak{S}_2(N-N')$ over pairs N, N' both lying in c, and $E_{\text{all}}[\mathfrak{S}_2]$ the same mean over all pairs in the band. Put

$$D_c \;:=\; E_{\text{same},c}[\mathfrak{S}_2] - E_{\text{all}}[\mathfrak{S}_2]. \tag{3}$$

Measurement 11 (the floor tracks D_c, including its shape)**.** At the octave $(2\cdot 10^6,\, 4\cdot 10^6]$:

depth	0	1	2	3	4	5
D_c	0.302138	0.046766	0.412504	1.052129	1.946109	3.171298
se_c	0.12400	0.04662	0.14834	0.24178	0.33253	0.43685

with correlation 0.9805 across depths. Neither row is monotone: both dip at depth 1, and that is the mechanism made visible. Depth 0 is the cell on which *none* of $3,5,7,11,13$ divides N, and excluding a class concentrates the shift just as fixing one does — if $3 \nmid N$ and $3 \nmid N'$ then both lie in $\{1,2\}$ mod 3 and $3 \mid h$ with probability $\frac{1}{2}$ against $\frac{1}{3}$ for a random pair. Both ends of the depth index concentrate h; depth 1 mixes the patterns and washes out. Depth is therefore a coarse index for D_c, and the floor follows the same coarse index (`lab_mask_placebo.py`).

Proposition 12 (the size mechanism is scale-invariant)**.** *D_c depends only on the distribution of the shift h in the residue classes mod $3,5,7,11,13$ that the cell fixes. That distribution is the same at every scale, so D_c is scale-invariant and predicts a fitted exponent of zero at every depth.*

Proof. For N, N' in a band, $\mathfrak{S}_2(N-N')$ depends on $N-N'$ only through the local factors at primes dividing $h = N-N'$. Depth does *not* fix the residue of N modulo each of $3,5,7,11,13$ — it fixes only how many of them divide N, so a depth-d cell is the union of the $\binom{5}{d}$ divisibility patterns of that size, and Measurement 11 records the mixing this causes. What the argument needs is weaker and does hold: each pattern fixes those five residue classes, so within a pattern the law of h modulo each of the five primes is determined; and the weights with which the patterns enter a depth cell are their densities among the even N of the band, which are the same at every scale. The induced law of h modulo those primes is therefore determined by the cell alone, and the band enters only through the residues of h at primes larger than 13, whose joint law over a band is the uniform one to $O(1/B)$, B the band length. Hence the difference (3) converges to a constant depending only on the cell, and its scale exponent is 0. ☐

Measurement 13 (the exponent, measured)**.** Measured over $(10^6, 2\cdot 10^6]$, $(2\cdot 10^6, 4\cdot 10^6]$ and $(4\cdot 10^6, 8\cdot 10^6]$ at 400000 sampled pairs per cell, the fitted exponents are

depth	0	1	2	3	4	5
e	+0.012633	+0.065546	+0.000936	+0.002904	+0.000241	−0.008428

— zero to three decimals at depths 2 to 5, and not at depths 0 and 1. Both exceptions are where D_c is small: the spread across the three octaves is about 0.004 at every depth while D_c itself runs from 0.045 to 3.17, so the relative spread the exponent reads is large only where D_c is. At ten times the sample the two move to -0.000879 and -0.016226, so depth 0 resolves to zero and depth 1 flips sign — the signature of noise rather than of a trend. The declared field's figures are the ones printed above; the ten-times-harder figures are a separate resolution and are not mixed into the table.

The sampling error of the pair average is 0.001358 at depths 0 to 4 and 0.000162 at depth 5, so at depth 1, where $D_c = 0.046766$ is six times smaller than any other, that error is the largest fraction of D_c anywhere in the table. Depth 1 is also the cell that carries no effect ($z = +1.11$ in Measurement 9), so nothing rests on it (`lab_cell_singular.py`).

Proposition 12 settles one thing and deliberately not another. Whatever produces the decay of the effect with N, it is not the arithmetic excess that produces the floor's size, since that excess does not decay at all. The decay exponent of the effect itself is not determined here: at the three shallowest depths the amplitude does not clear the exact floor at any scale measured, so no exponent can be fitted there.

6 Summary

- Lemma 1: the fluctuation of any cell mean under the sign null is exactly $Q_{cc}/n_c^2 - 2Q_{ca}/(n_c n) + Q_{aa}/n^2$; all three terms are needed and all three are one convolution each.

- Observation 4 and Note 6: that quantity decays like $(\log N)^{-1/2}$, not $n_c^{-1/2}$. A count-based interval is a factor 9.3 too narrow across a factor 128 in N, and absolutely too narrow by factors of 7 to 158 at the top octave, the largest of them at the shallowest cell.

- Lemma 7 and Measurement 9: the cell effect survives label permutation; the floor does not. The floor is a second measurement of the same correspondence, and quoting against it is conservative by 3.8 at the depth that carries the effect — two orders is the depth that carries none.

- Proposition 12: the quantity that predicts the floor's size is scale-invariant, so it is not what makes the effect decay.

- The mechanism — coherence of $u_c(v)$ across the cell — is generic. Any cell mean of a field whose summands share a common arithmetic input has the same property.

References

[1] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References here are to the arXiv version, which we have read, and not to the published version, which we have not.

[2] *The exact second moment of* $\sum_{n<N} \Lambda(n)\mu(N-n)$, *and why Chowla's conjecture does not control it*, companion paper.

A Reproducibility

Script	Result file	Supports
`lab_cellmom_montecarlo.py`	`lab_cellmom_montecarlo.txt`	Measurement 3
`lab_cell_floor.py`	`lab_cell_floor.txt`	Measurement 5
`lab_mask_placebo.py`	`lab_mask_placebo.txt`	Measurements 9, 11
`lab_cell_singular.py`	`lab_cell_singular.txt`	Measurement 13

Lemma 1, Lemma 7 and Proposition 12 are proved and use no computation. The measurements state the band, the cell index and the statistic they were taken of; each was run with its decision rule fixed in advance, and each cell-indexed claim was run against the permutation control of Lemma 7.